

CineByte : Film Discovery and Personal Tracking Platform

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Abstract

Film fans today use multiple tools to track, review and watch movies. This creates friction — users switch between apps to log a film and find its trailer. CineByte is a web platform that solves this problem by combining tracking movie and media - Youtube embeds- all in one place. The platform lets users log films, write reviews, rate content and watch trailers without leaving the site. This study documents the analysis and design phase of CineByte, covering audience profiling, content inventory and system architecture. The result is a responsive, social film platform that reduces user effort and have potential for a cleaner, more connected viewing experience.

1. Introduction

People who love films often face the same problem. They use one app to find a trailer, another to log what they watched, and a third to read or write reviews. This breaks the experience into separate steps across separate tools. The user must leave one platform, open another, and then return — just to complete a single task.

Platforms like Letterboxd allow users to log films and write reviews. They work well for record-keeping. However, they do not support embedded media. Users cannot watch a trailer or video review inside the platform. This forces them to open YouTube in a new tab, watch the content, and come back to log it. That extra step adds friction to what should be a simple task.

YouTube, on the other hand, hosts a large library of trailers and video reviews. But its UI is built for content discovery, not personal tracking. Users can save videos to a "Watch Later" list, but the tool offers no structured logging, no rating system, and no community review features.

CineByte addresses this gap. It is a film discovery and social tracking platform that brings together movie logs, user reviews, star ratings, and embedded YouTube content in one responsive layout. Users can watch a trailer and log the film in the same session, on the same page. The platform draws on the core strengths of existing tools while removing the steps that break the user flow.

This paper covers the analysis phase for CineByte. Section 2 defines the problem statement. Section 3 states the project objectives. Section 4 profiles the target audience. Section 5 provides a competitive benchmark. Section 6 outlines the content inventory and system modules. Section 7 presents the expected result.

2. Problem Statement

Current media-tracking platforms, such as Letterboxd, mainly focus on feature films and traditional cinema. This creates a gap for users who frequently consume other types of digital content, especially episodic shows and Youtube videos, as there are no dedicated platforms for them to log, review, and organize their viewing experience.

Although YouTube provides features like the “Watch Later” list, it is limited to personal use and does not support social interaction, structured logging, or community-based reviews. As a result, users are unable to share their viewing activities or engage in discussions in a meaningful and organized way.

This project aims to address this gap by developing a platform that allows users to track shows and YouTube videos in one place. A key feature of the system is the ability to embed YouTube links directly into user logs, creating a seamless “watch and record” experience. By combining content tracking with social interaction features such as reviews and ratings, the platform offers a unique service tailored to modern digital content consumption.

3. Objectives

(i) To develop a platform that allows users to log, track and organize movies, episodic shows and YouTube videos in a structured way in a single system.

(ii) To provide a feature that enables users to embed YouTube links directly into their viewing logs, creating a seamless watch-and-record experience.

(iii) To integrate social interaction features such as reviews, ratings and discussions so users can share and engage with others about the content they watch.

4. Audience Profiling

The target audience for this platform consists mainly of Gen Z and Millennials, aged between 15 to 35 years old. This group represents the largest segment of users who actively consume YouTube content, web series and other forms of digital media.

The platform is designed to be inclusive of all genders, focusing on individuals who regularly engage with online content and enjoy sharing their opinions and experiences with others. In terms of socioeconomic background, the primary users are students and young professionals who have consistent access to high-speed internet and personal digital devices such as smartphones, tablets or laptops.

In terms of technical proficiency, the target audience is considered highly skilled. As digital natives, they are familiar with platforms like YouTube and social media applications. They are comfortable using features such as search functions, managing personal profiles and interacting through ratings, comments and reviews. Therefore, the platform should provide an intuitive and interactive interface that supports these user behaviors.

5. Competitive Benchmarking

A competitive benchmarking analysis was conducted to evaluate two existing platforms, Letterboxd and Serializd. The purpose of this analysis is to identify their strengths and weaknesses in terms of usability, features, and user engagement, in order to inform the design and development of the proposed website

Table 1 competitive benchmarking analysis table

Criteria	Letterboxd	Serializd	Academic Reasoning	Source

User Interface (UI/UX)	Simple UI but described by some as cluttered or having sensory overload due to many options in a single pop-up window.	Newer, clean-UI option; mimics the aesthetic and social structure of Letterboxd but with free custom colors.	Information Architecture (IA) principles suggest that structured hierarchy and clear labeling are essential to enhance navigation and ensure content clarity for an efficient user experience.	1,2
Features (logging, reviews, ratings, lists)	Focuses on film logging, 5-star nuanced rating scale, list creation, and a social diary; limited TV tracking, mostly 1-season shows.	Dedicated to TV tracking with episode-level logging, reviews, lists, and a unique badge system for accomplishments.	Systematic evaluation of usability focuses on task completion and efficiency; features like detailed logging allow users to develop mental models of the information space.	3,4
Content Focus	Exclusively focused on cinema and films; cultivated a base of highly-engaged cinephiles.	Focused exclusively on television shows, including series and episode-level format tracking.	Cognitive Load Theory says that when content is too complex or hard to understand, it can make people less interested and think it's not as good, even if the content itself is actually good.	1, 5, 6
Community and Social Interaction	Highly social with followers and viral-style reviews; community can be competitive and sometimes focuses more on performing criticism.	Passionate community that is reactive and small, facilitating a closer feel; social features include following friends and sharing lists on home pages.	Digital media studies indicate that the endowment effect and psychological ownership enhance valuation and post-consumption satisfaction through communal engagement.	1, 6
Ease of Use	Considered simple for seasoned users; includes comprehensive search by genre, release year, and crew.	Manual-entry platform requiring users to find episodes by hand; mobile website reported to have overflowing UI elements.	The 'Three-Click Rule' and 'Intuitiveness' are standard IA criteria used to evaluate if users can access information with minimal effort.	3
Limitations	Customization options like stats and poster changes are only available if you pay a yearly fee, which ranges from \$19 to \$49.	Reported as bugged (glitched stats/ratings) and developed by only one person; lacks an API for	The 'Zero-Price Effect' suggests that while free access boosts perceived value, it may inversely affect engagement if users feel less	1, 6

third-party
integrations.

'ownership' over the
platform.

6. Content Inventory and System Modules

The content inventory outlines all digital assets required in the system, including text, images, icons, and video elements. These contents are essential to support different features of the website.

Table 2 Content Inventory

No	Page/Module	Content Type	Description	Source
1	Homepage	Text	Website introduction and	Self-written
2	Homepage	Image	Banner or background design	Canva
3	Movie Tracking	Image	Movie posters	TMDb API
4	Movie Tracking	Text	Movie titles and descriptions	API
5	YouTube Tracking	Video	Embedded YouTube videos	YouTube
6	Profile	Text	User information	User input
7	Review Section	Text	User reviews and comments	User input
8	Rating System	Icon	Star rating icons	FontAwesome
9	Navigation Bar	Icon	Menu icons (Home, Profile, Search)	Flaticon

In addition to content inventory, the system requires various digital assets to support the visual design and functionality of the website. These assets include images, icons, videos, and graphical user interface elements that enhance user experience and interaction.

Table 3 Digital Asset List

No	Asset Name	Type	Purpose	Source	Format
1	Website Logo	Image	Branding of website	Canva	PNG
2	Navigation Icons	Icon	Menu navigation (Home, Profile, Search)	Flaticon	SVG
3	Movie Posters	Image	Display movies in list	TMDb API	JPG
4	YouTube Thumbnails	Image	Preview of videos	YouTube	JPG
5	YouTube Videos	Video	Embedded video tracking	YouTube Embed	URL
6	Rating Stars	Icon	User rating system	FontAwesome	SVG
7	Profile Avatar	Image	User profile picture	User Upload	PNG/JPG
8	Background Banner	Image	Homepage design	Canva	JPG
9	Buttons UI	Graphic	Interaction elements	CSS / Design	-
10	Review Text	Text	User comments	User Input	-

System Modules

The system is divided into several modules, each handling specific functionalities:

1. User Management Module

This module manages user registration, login and profile information. It allows users to create accounts and personalize their profiles.

2. Movie Tracking Module

This module enables users to add, view, and manage a list of movies they have watched. It displays movie details such as title, description, and poster.

3. YouTube Tracking Module

This module allows users to track YouTube videos they have watched by embedding video links into the system.

4. Review and Rating Module

Users can write reviews and rate movies or videos using a rating system.

5. Search and Navigation Module

This module helps users search for movies or videos and navigate easily between different pages of the system.

The system is structured into several key modules, including user management, movie tracking, YouTube tracking, review and rating, and search and navigation, where each module is responsible for specific functionalities that support the overall operation of the website. The content inventory complements these modules by identifying and organizing all required content such as text, images, icons, and video elements, ensuring that each page and feature has the necessary information to function effectively. In addition, digital assets such as logos, navigation icons, movie posters, banners, rating icons, and embedded videos are carefully selected and managed to support the system's visual and interactive components. The integration of system modules, content inventory, and digital assets ensures a well-structured, consistent, and efficient platform that enhances usability, improves user experience, and provides a scalable foundation for future system development.

7. Expected Result

The development of this platform aims to fix the limitations of current media-tracking systems like Letterboxd and YouTube's basic "Watch Later" feature. The system will provide users with a single space to log, track, and organize episodic shows and YouTube videos in a clear and meaningful way. The goal is to create a fully functional platform that supports YouTube link embedding. This will let users easily record what they watch while enjoying a smooth "watch-and-log" experience. This approach will improve convenience and cut down on the disorganization that comes with tracking various digital content. Moreover, the platform should promote active user participation through reviews, ratings, and discussions. This will create a more social and interactive atmosphere compared to current tools. Users will be able to share their viewing experiences, find new content through community interaction, and take part in structured conversations. In summary, the expected result is a modern hybrid content-tracking platform that connects traditional film logging with digital content consumption. It should also lead to higher user satisfaction, better content discovery, and support future research into personalized recommendation systems and social viewing analytics.

References

- [1] [1] Achriom vs Every Media Tracker: A Complete Comparison (*Achriom vs Every Media Tracker: A Complete Comparison*, n.d.) <https://www.achriom.com/compare/>

- [2] [2] I Tried Binge, the Letterboxd Alternative That I Now Like More Than Letterboxd – Lifehacker (*I Tried Binge, the Letterboxd Alternative That I Now Like More Than Letterboxd* / Lifehacker, n.d.)
<https://lifehacker.com/tech/i-tried-binge-the-letterboxd-alternative>
- [3] [3] Albadi, H. B. (2025). Information Architecture of King Abdullah University of Science and Technology and Johns Hopkins University Websites: Comparative Analysis. *American Academic & Scholarly Research Journal*, 14(10). <https://aasrc.org/aasrj/index.php/aasrj/article/view/2300>
- [4] [4] El-firjani, N.F., Elberkawi, E.K., & Maatuk, A.M. (2017). A Method for Website Usability Evaluation : A Comparative Analysis. *International Journal of Web & Semantic Technology*, 8, 01-11.
<https://aircconline.com/ijwest/V8N3/8317ijwest01.pdf>
- [5] [5] (*IMDB vs. Letterboxd: What's the Difference? A Statistical Analysis*, n.d.)
<https://www.statsignificant.com/p/imdb-vs-letterboxd-whats-the-difference>
- [6] [6] Mudambi, M., & Wang, Y. (2026). Free to Watch, Harder to Love: Subscription Effects on Film Evaluation. *Harder to Love: Subscription Effects on Film Evaluation* (January 27, 2026).
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6141886